Smaran Vallabhaneni

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Education

Olin College of Engineering — Needham, MA
Bachelor of Science, Engineering in Computing (GPA: 4.0) — *Aug. 2025 – May 2029*
- **Relevant Courses:** Modeling & Simulation, Linear Algebra, Data Science, Multi-Variable Calculus, Software Design (Python), Kernel Development (C), Products & Markets, Introduction to Sensors, Instrumentation & Measurement

Skills

Programming Languages: Java, C/C++, Python, pandas, HTML, CSS, JavaScript, Dart, LaTeX, MySQL, C#, R, PBASIC, Bash, Zsh, PowerShell, Swift, MATLAB
Operating Systems / SBCs: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, and Arch based), Raspberry Pi, Arduino
Tools: Blender, Fusion360, AutoCAD, Autodesk, SolidWorks, Unity, Figma, Adobe Creative Suite, Excel, Git, IntelliJ, VSCode, Xcode, Overleaf, Node.js, GPL Cver, Simulink, MATLAB, Tableau Desktop, Docker, MissionPlanner, ArduPilot, GitHub Copilot, Claude Code
Frameworks: LLMs, RAG Systems, Django, PyTorch, Kivy, Flutter, SwiftUI, React, AngularJS, .NET, Ollama, System Design, Metal API, ROS/MAVROS, OpenCV, MediaPipe, Blender Python API
Other: UI/UX, CI/CD Pipelines, ESP32, Computer Vision, Internet of Things, AWS, GCP, Azure, Dev Ops, HuggingFace, NLP, Cryptography, Encryption, Databases, Database Management, Statistical Analysis, Version Control, Embedded Systems, Algorithm Development, Data Structures, Object-Oriented Programming, Testing and Debugging
Languages: English, Spanish, French, Chinese, German (Limited Working), Telugu

Projects

CLOS (Command Line Operating System) | *C++, Java, HTML, CSS, Bash* — Apr. 2023 – Present
- Building a cross-platform centralized terminal and graphical math tool with equation solving, graphing, symbolic calculus, and trigonometry.
- Engineered a custom database engine, AES encryption module, symbolic math parser, and a 3D rendering library from scratch in Java; achieved consistent 60 FPS graphic rendering and executed complex multi-variable calculus computations under 50 ms.

ClassicTunes | *Swift, SwiftUI, Metal API* — Oct. 2025 – Present
- Developing an open-source modern remake of iTunes 7–10 for Apple Silicon macOS, faithfully recreating CoverFlow, MiniPlayer, and Smart/Genius Playlists.
- Leveraged Apple's Metal API to build a custom GPU-accelerated CoverFlow engine, matching the performance of the native Apple Music client with a fluid 120 FPS render rate across 150+ albums on Apple Silicon.

Whoa-Scope | *Python, Kivy* — Feb. 2026 – Present
- Co-developed an open-source oscilloscope fork to resolve compatibility issues and desktop UX flaws, gaining organic adoption from 50% of the class (40 students) and faculty recognition.
- Profiled and refactored memory allocation, slashing RAM consumption by 200+ MB to maximize stability and prevent lag on lab computers.

3D Gesture Tracking | *Blender, Python, OpenCV* — Apr. 2026
- Engineered a real-time computer vision add-on published on Blender Addons, enabling mouse-free viewport manipulation (pan, zoom, rotate) via hand gesture tracking.

Experience

Olin College AERO Project Team — Needham, MA
Head of Ground Station Development — Software Engineer — *Sep. 2025 – Present*
- Leading software development for an autonomous fixed-wing aircraft competing in the AUVSI SUAS competition, targeting precision water delivery to GPS-specified targets using image recognition.
- Architected ground station software integrating ArduPilot, MissionPlanner, and ROS/MAVROS systems to bridge an on-board NVIDIA Jetson computer with ground controls.
- Engineered a robust fly-by-wire telemetry pipeline, achieving a consistent sub-100ms round-trip latency for real-time sensor monitoring and safety interlocks.

Geeky Insights — Remote
Full-Stack Android App Developer — Internship — *Apr. 2024 – Apr. 2025*
- Designed end-to-end UI/UX of a customizable audio alarm clock app using Figma and Adobe Creative Suite, prioritizing accessibility and intuitive design for a diverse user base.
- Engineered an end-to-end frontend/backend architecture using Flutter, Dart, and Java, optimizing the compilation pipeline into a lightweight, bug-free 15 MB production package.
- Optimized application memory layout to support low-resource devices running Android 9+, reducing RAM consumption to match the baseline footprint of Android's native system clock.

Activities

Olin ICPC — Needham, MA
Competitive Programmer — Team Member — *Jan. 2026 – Present*
- Solving high-difficulty algorithmic problems under competitive time constraints in bi-weekly ICPC-style contests.
- Applying advanced techniques including segment/Fenwick trees, dynamic programming, graph algorithms, string algorithms, network flow, and computational geometry.

Research & Publications

C++ and Java Performance in Android Environments — Aug. 2025
- Benchmarked C++ and Java runtime performance across multiple Android devices; demonstrated C++ advantages in resource-intensive workloads and Java advantages in stability and memory safety.

C++ Memory Safety and Integrity Analysis — Oct. 2024
- Analyzed C++ memory safety by constructing targeted leak scenarios and comparing behavior against Java equivalents; findings informed recommendations for language selection in safety-critical mobile systems.